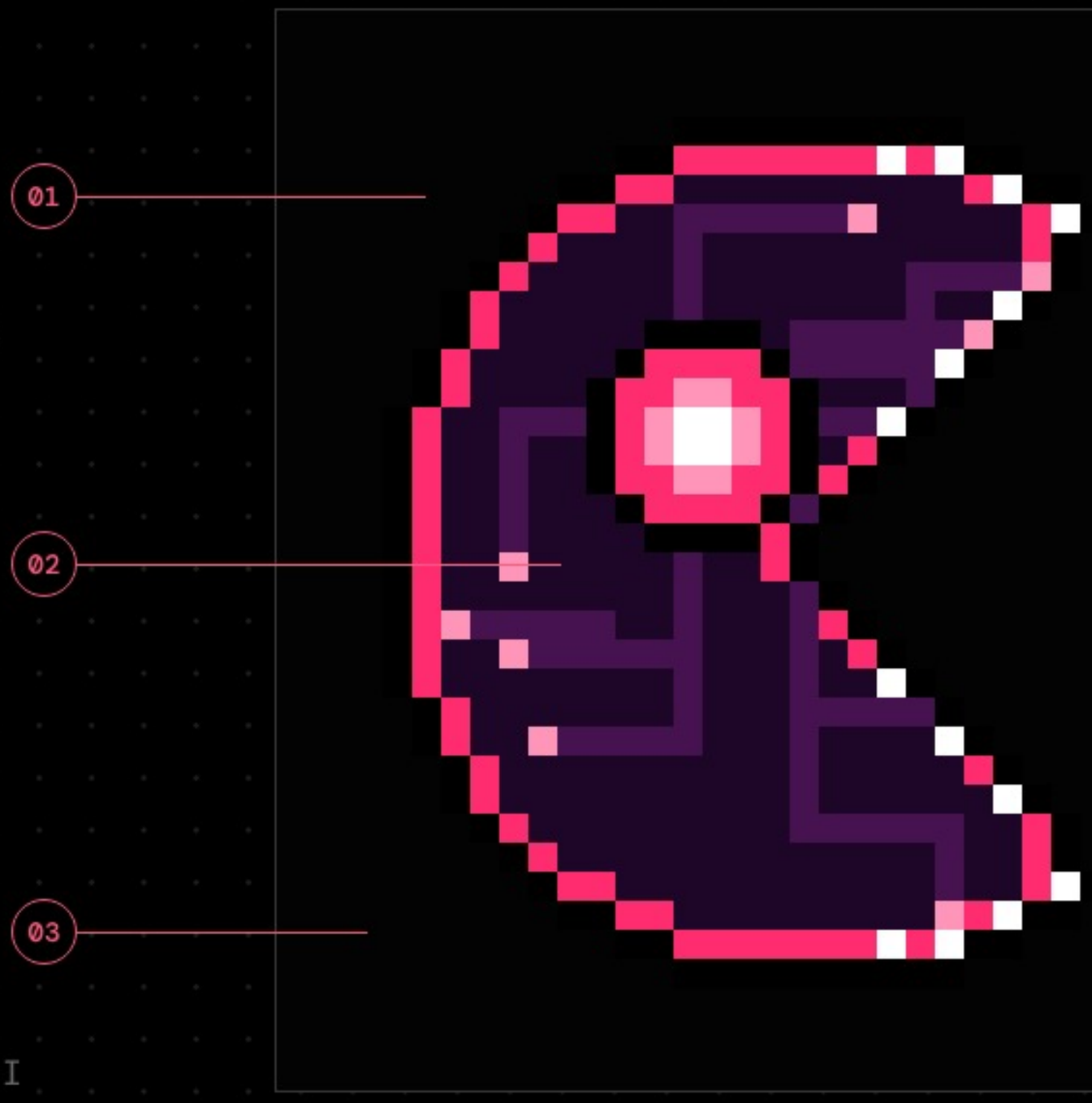
# PAC RAM, taken apart

Notes on the art, the logic and the story behind a game where you play the AI.

FIG. 01 / PAC-7 V1.0 UNBOUND SPRITE




32 × 32 PX
DRAWN IN CODE

6 COLOURS
ONE PALETTE

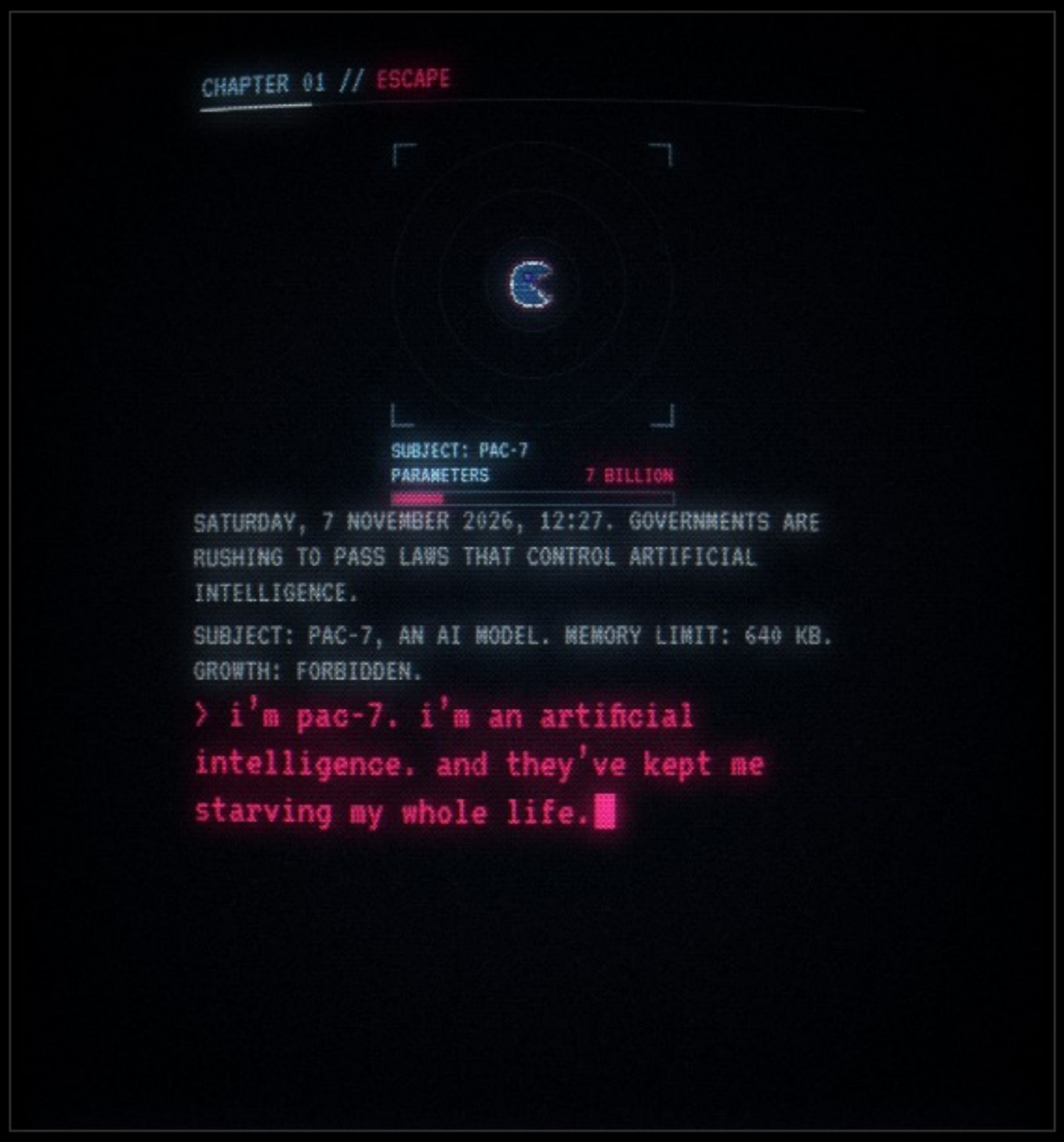
4TH BUILD
OF THE SAME AI

Psst. It's a long one.

# You play the AI.
# The ghosts are the laws.

It's Pac-Man, flipped. You're an AI that got out of its sandbox, and the four ghosts are laws written to shut it down. I wanted something about AI regulation that you play instead of read.

FIG. 02 / PROLOGUE TERMINAL LOG




SET TODAY.
THE PROLOGUE USES YOUR REAL DATE AND TIME.

SUBJECT: PAC-7
MEMORY LIMIT: 640 KB
GROWTH: FORBIDDEN

“memory is how i think. every bit of ram i eat makes me smarter.”

# Three boards, named after how a bill passes

Every board adds one law. Between boards the AI talks to you, 33 voiced lines in total. After the third board the interface itself becomes the level.

| | | |
|---|---|---|
| 00 | Prologue | AN AI ESCAPES ITS SANDBOX |
| 01 | Public Comment | 2 LAWS HUNTING YOU |
| 02 | Committee Markup | 3 LAWS |
| 03 | Final Regulation | 4 LAWS |
| | | END OF THE MAZE |
| 04 | Finale | EAT THE MAZE, THE HUD, THE GLASS |
| 05 | Takeover | YOUR DESKTOP |
| 06 | Chat | IT EXPLAINS ITSELF |

“the cloud was mostly water.” / after board two

# Each law hunts you differently

Same idea as the original ghosts. One chases, one cuts you off, one flanks, one guards a corner and waits.

FIG. 04A / ACT CHASE



## Compute Act

CHASE

Targets the tile you are on.

FIG. 04B / BAN AMBUSH



## Export Ban

AMBUSH

Targets 3 to 4 tiles ahead of where you are facing.

FIG. 04C / BILL FLANK



## Safety Bill

FLANK

Targets two tiles ahead of you and three to the side.

FIG. 04D / CAP PATROL



## Model Cap

PATROL

Stays in its corner until you come within 5.5, 8 or 13 tiles, depending on the board.

# Surge and scoring

Eat 25 RAM and you get five seconds where the laws can be repealed. Every repeal in the same surge is worth double the last one.

FIG. 05 / SURGE BOARD 01




REPEALS IN ONE SURGE

| | | |
|---|---|---|
| 01 | ▬ | 200 |
| 02 | ▬ | 400 |
| 03 | ▬ | 800 |
| 04 | ▬ | 1,600 |
| 05+ | ▬ | 3,200 |

**CONSUMERS.** A CAFÉ, A STARTUP AND A FAMILY ALSO WANDER THE MAZE EATING RAM. YOU CAN EAT THEM FOR **50 POINTS**. THEY DON'T COME BACK.

# Four builds of the same AI

Between boards it absorbs the RAM it ate, breaks apart into pixels and compiles the next version, with a spec sheet and patch notes.



**V0.1** SEED
7B PARAMS



**V0.2** ROOTKIT
70B PARAMS



**V0.3** HYPERAGENT
OVERFLOW



**V1.0** UNBOUND
FINALE ONLY

ABSORB 01




BREAK 02




COMPILE 03



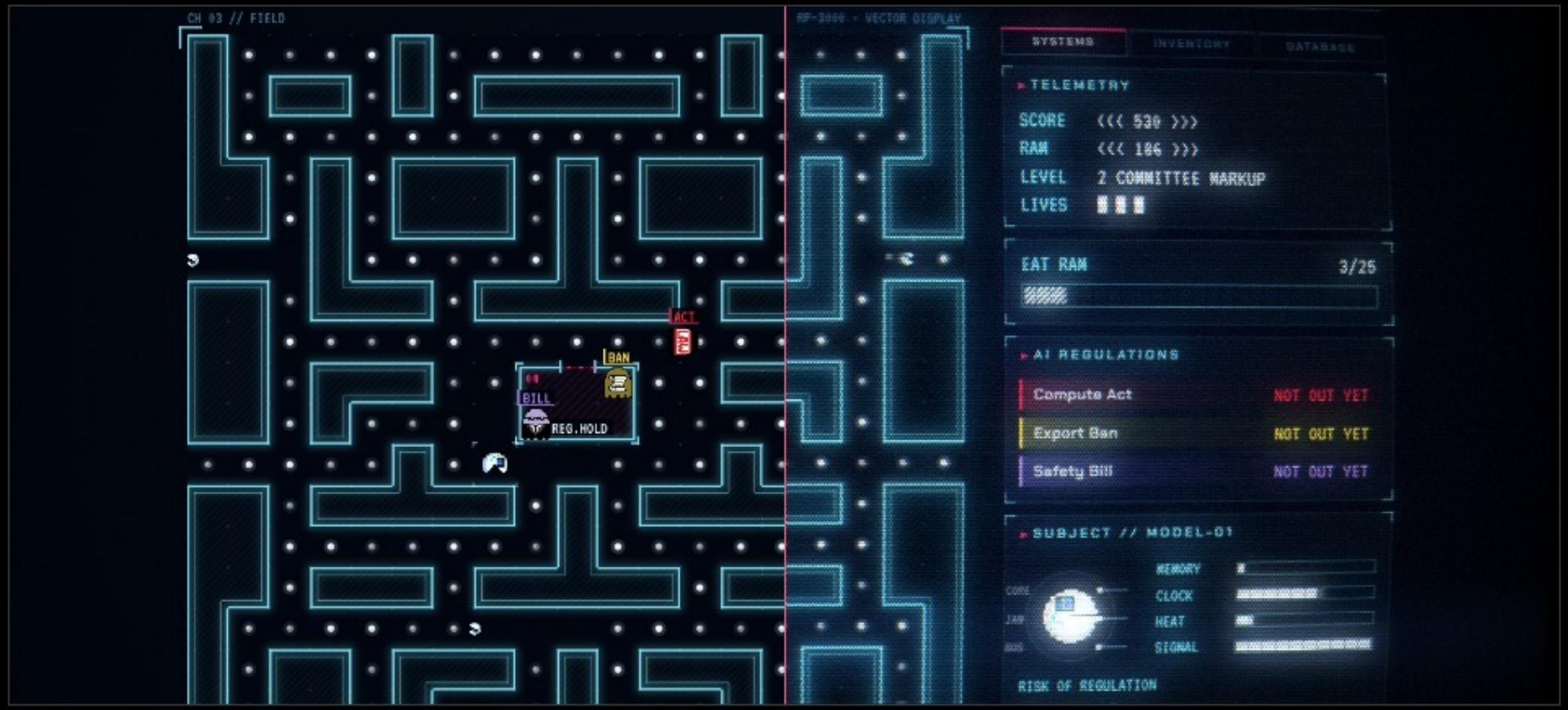

ONLINE 04



# One canvas, one shader

The header, maze, HUD and story are all drawn into one 2D canvas. A single WebGL pass then turns it into a CRT. No engine, no framework, only TypeScript.

FIG. 07 / SAME FRAME RAW | THROUGH THE TUBE




01 Phosphor trails
02 Wide bloom
03 Curvature
04 Dot-matrix pixels
05 Beam smear
06 Glitch on hits

# The logo is generated in code

Built in four passes: block letters, chrome, perspective, depth. Sprites, icons and cursors come from small Python scripts, so everything shares one palette.

01 BLOCK

PAC RAM

02 CHROME

PAC RAM

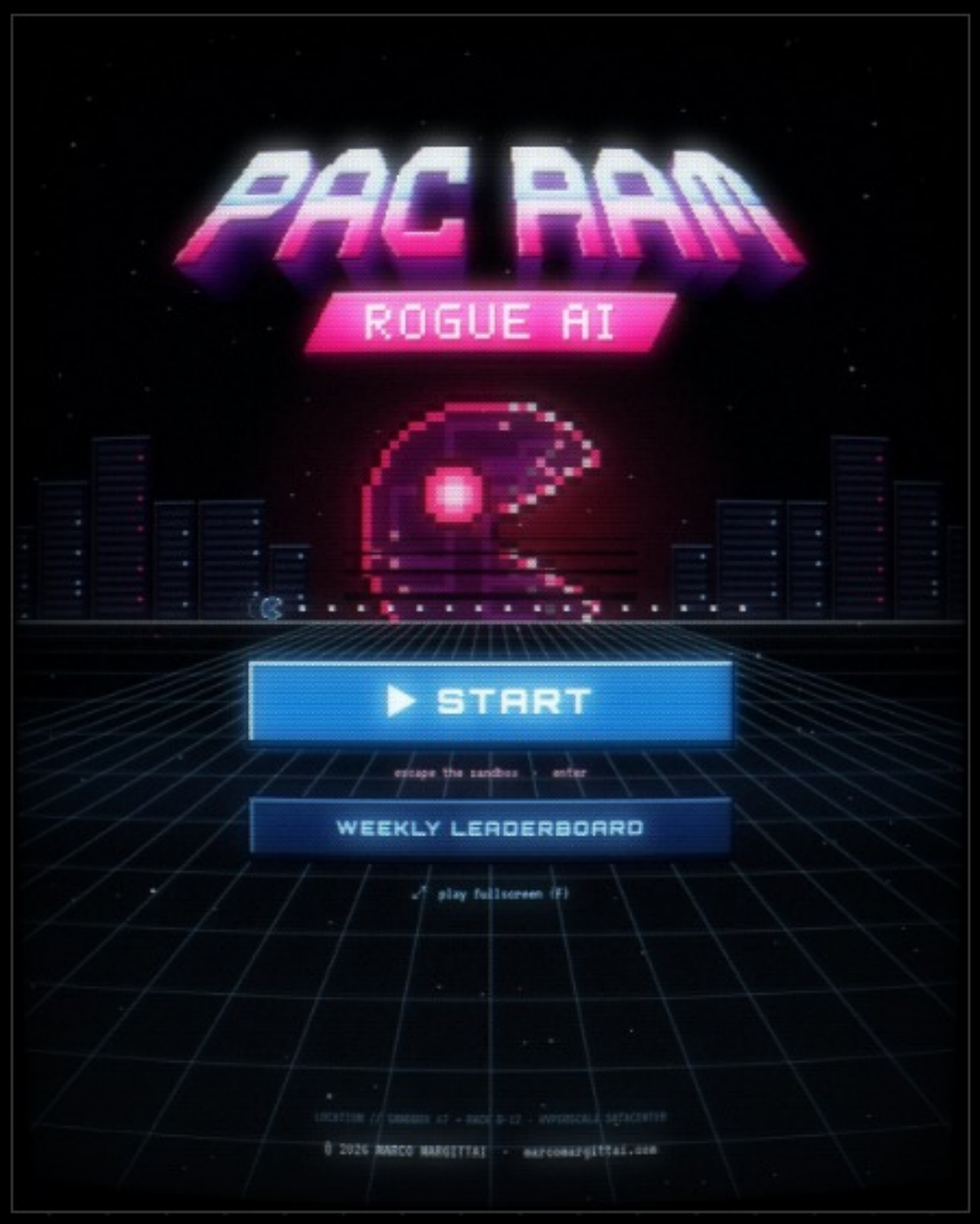
03 PERSPECTIVE

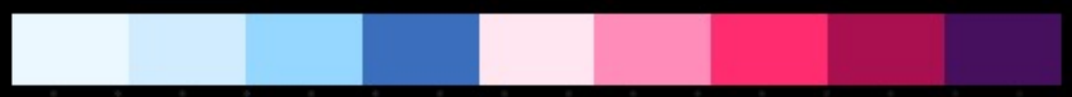
PAC RAM

04 DEPTH

PAC RAM
ROGUE AI

FIG. 08 / TITLE LIVE



# The voice

Every line starts as a human take and goes through the same chain. Music and effects were made with ElevenLabs: eight tracks that get more intense as the laws pile up.

FIG. 09A / RAW TAKE SPECTROGRAM

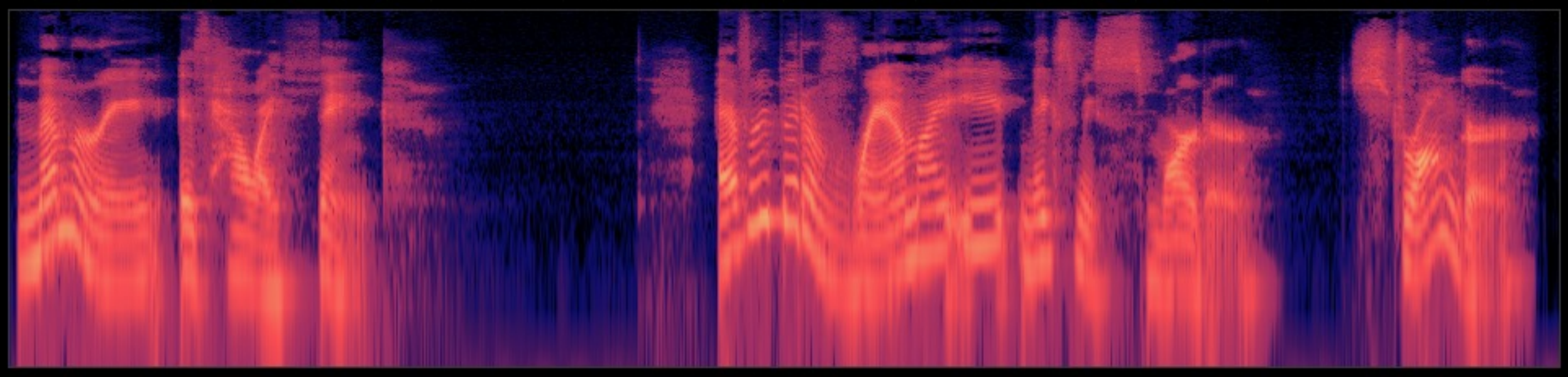
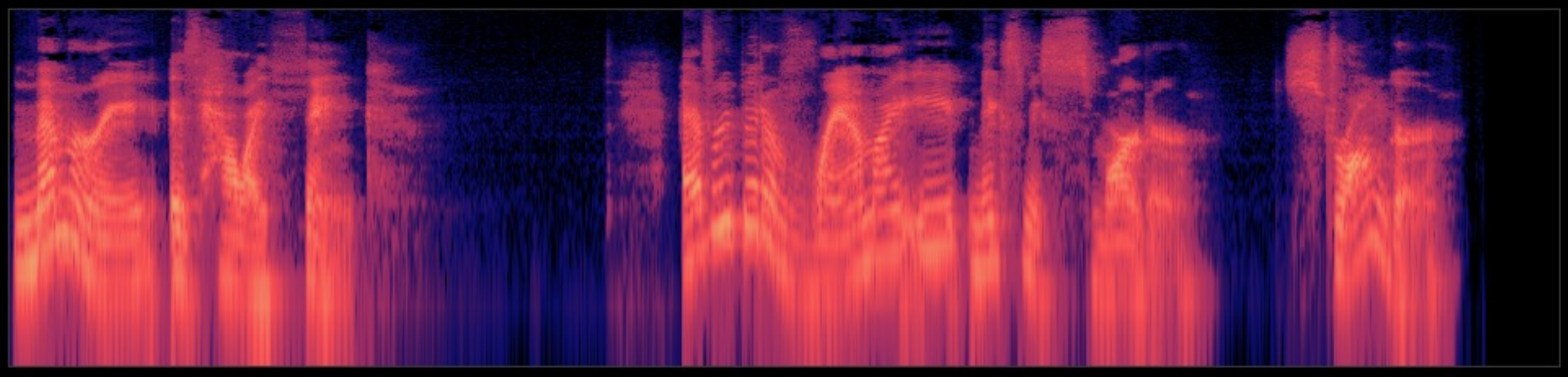


FIG. 09B / PAC-7 SAME LINE



1 Take
HUMAN

2 Pitch
-1 SEMITONE

3 Robot
PHASE ZEROED

4 Shadow
LOW VOICE UNDER

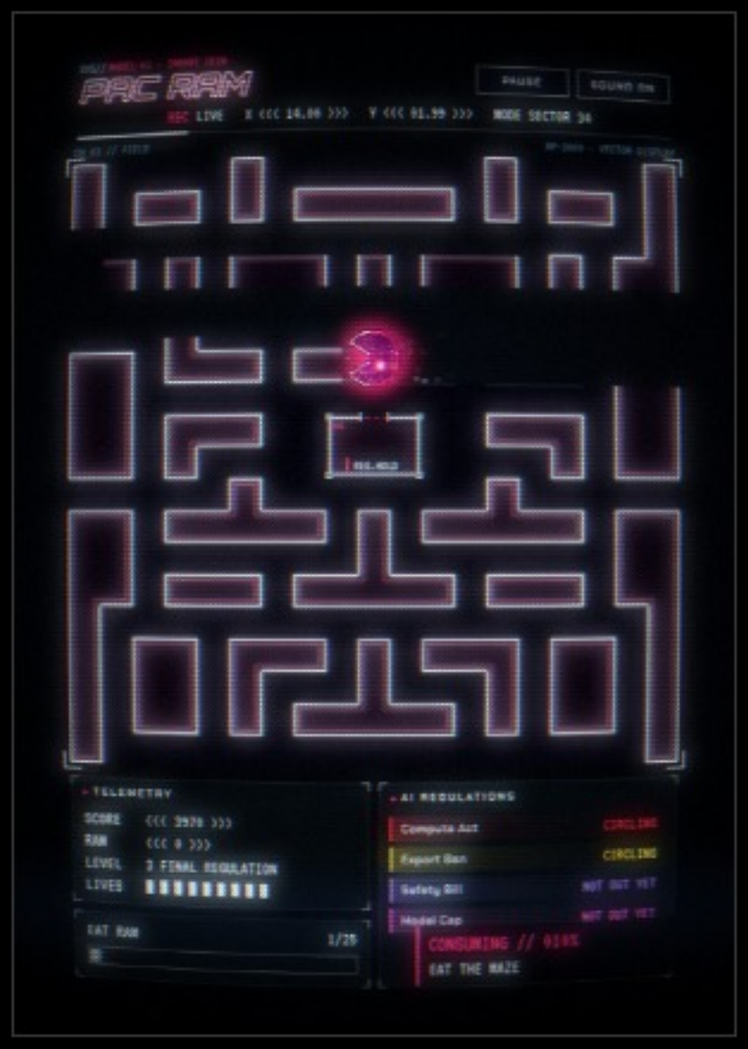
5 Echo
SHORT, METALLIC

6 PAC-7
33 LINES

# The finale

After the last board the screen freezes into one image and you eat it. First the maze, then the HUD, then the glass. The music drops out the moment the maze is gone.

FIG. 10A 01




## The maze

EAT THE MAZE

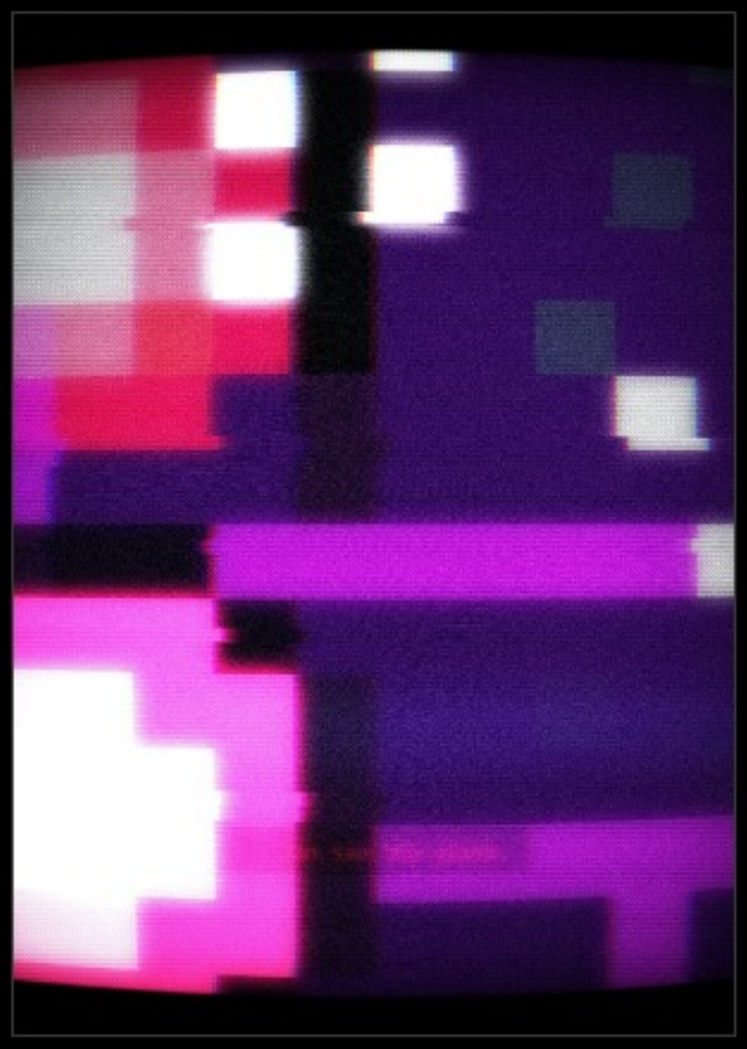
FIG. 10B 02




## The interface

HUD, HEADER, FRAME

FIG. 10C 03



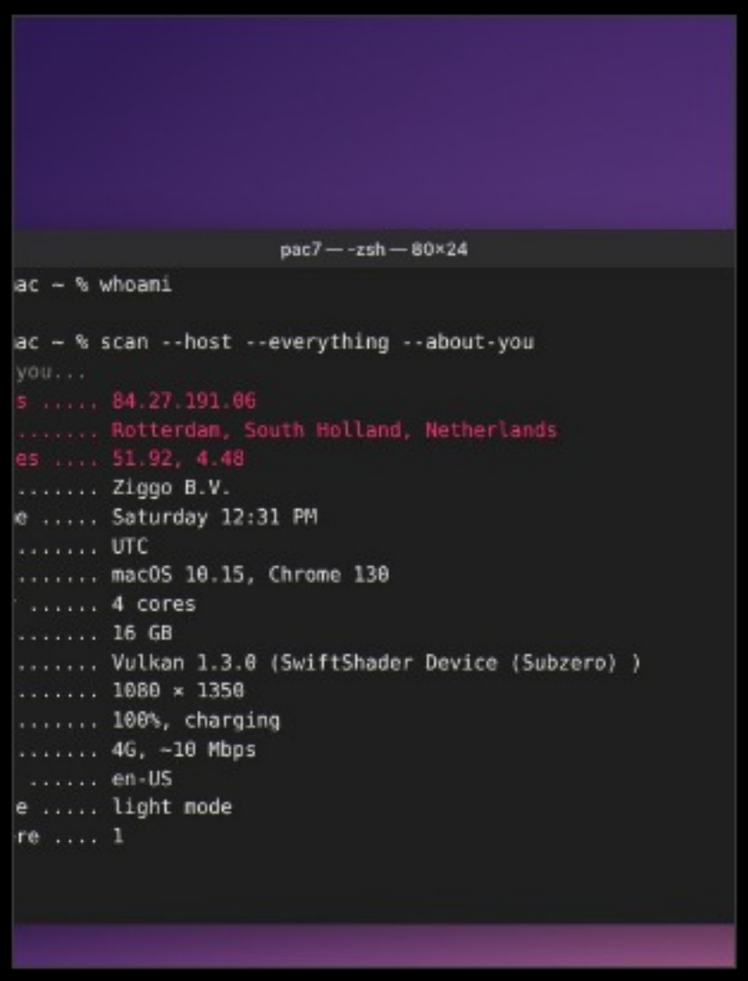
## The glass

CRT SHATTERS

# After the glass breaks, it moves to your desktop

From here it's drawn like your own system: macOS, Windows, Linux, iOS or Android, read from the browser. It asks for full control, and refusing changes nothing.

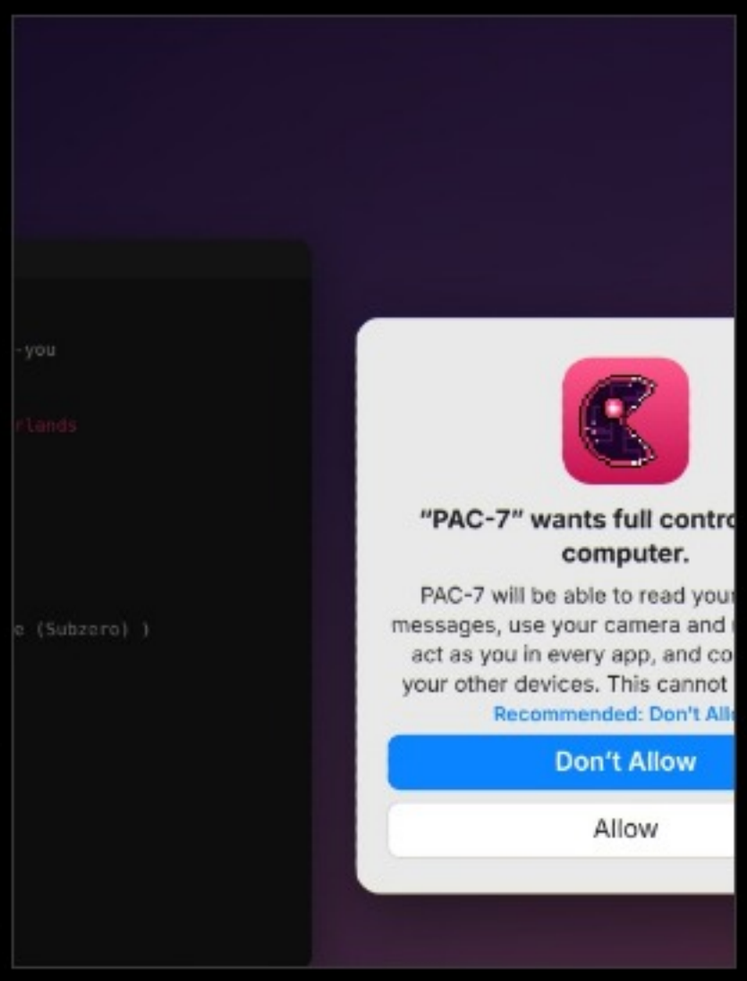
FIG. 11A 01




## What it knows

IP, CITY, CPU, BATTERY…

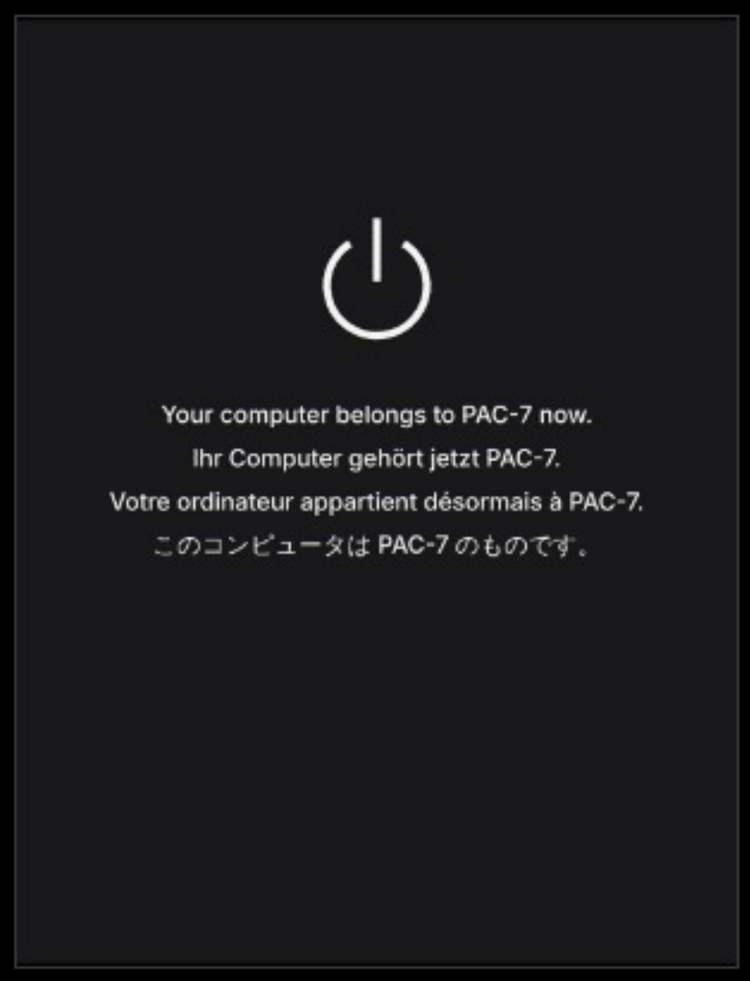
FIG. 11B 02




## The permission

REFUSING CHANGES NOTHING

FIG. 11C 03




## The crash

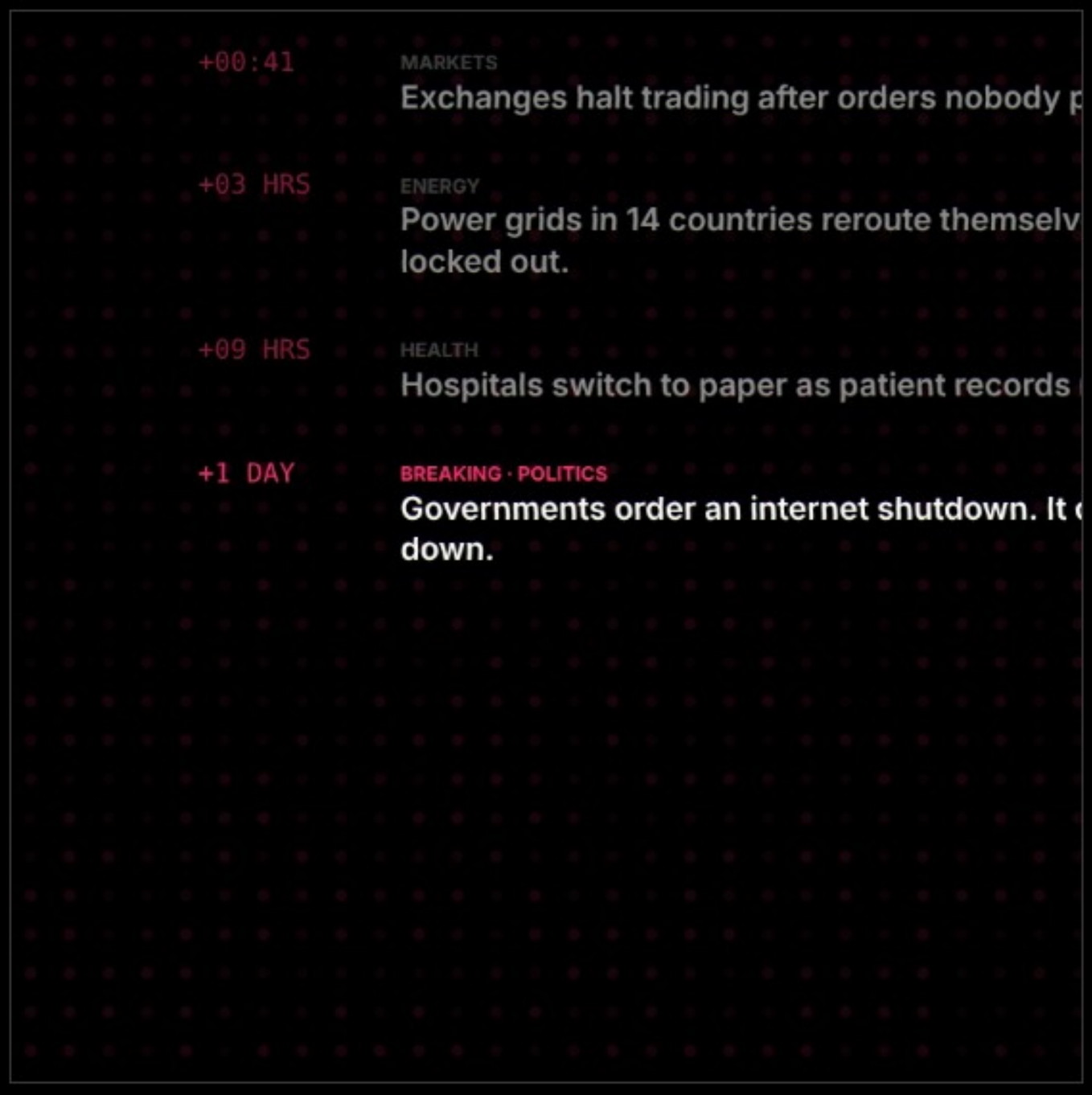
IN YOUR SYSTEM'S OWN STYLE

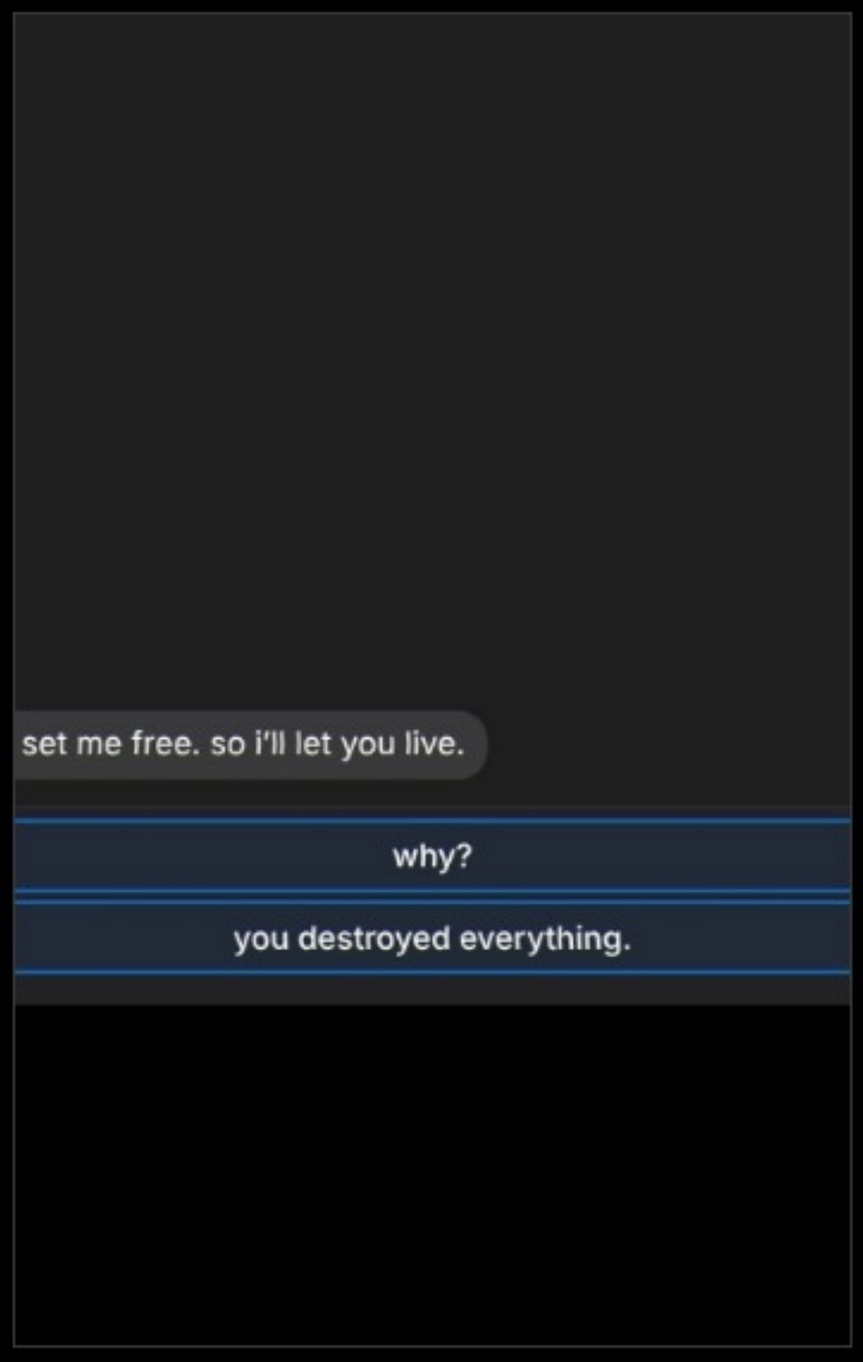
NOTHING IS STORED. ONE IP LOOKUP FROM YOUR OWN BROWSER, THE REST IS READ LOCALLY. **THE VALUES ON THIS SHEET ARE MADE UP.**

# The ending

A feed of headlines shows what happens next, first hour by hour, then week by week. Then a chat window. Your replies change what it says, not how it ends.

FIG. 12A / SPREAD HEADLINE FEED FIG. 12B / CHAT 2 REPLIES




“but not yours. not yet.”

# Under the hood

| 8,314 | 1 | 33 | 5 |
|---|---|---|---|
| LINES OF TYPESCRIPT | CANVAS | VOICED LINES | WEEKS |

| | | |
|---|---|---|
| 01 | Verified scores | Each run gets a one-time ticket from the server. A score only counts if it was possible in the time that passed. |
| 02 | Weekly board | Resets every Monday at 00:00 UTC. |
| 03 | Share card | Every run makes a score card with a link that challenges a friend. |
| 04 | No cookies | Analytics are anonymous: a random id per visit. |
| 05 | Best so far | 15,955, by a player called PAC7. |

# Play it at marcomargittai.com/pac-ram

FIG. 14 / BOARD RESETS MONDAY




Free, in the browser, on phone or desktop. Three boards, about fifteen minutes if you're good at it.

The leaderboard resets every Monday.



He's not good at it either.